THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL UNIT TEST SET 2

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If $x + \frac{1}{x} = 3$, then $x^2 + \frac{1}{x^2} =$ 1

(a) 9 **(b)** 7 **(c)** 8 **(d)** 6

2. If $ab > 0$ and $a + b < 0$, the point (a, b) lies in: 1

(a) Quadrant I **(b)** Quadrant III **(c)** Quadrant II **(d)** Quadrant IV

3. If $p(x) = 5x + 2$ and $q(x) = 3x - 7$, then $p(x) + q(x) =$ 1

(a) $8x - 4$ **(b)** $8x - 5$ **(c)** $8x - 6$ **(d)** $9x - 5$

4. $\sqrt{3} \times \sqrt{12} =$ 1

(a) 8 **(b)** 5 **(c)** 7 **(d)** 6

5. The value of 1003^2 is: 1

(a) 1006011 **(b)** 1006008 **(c)** 1006009 **(d)** 1006010

6. The perimeter of the triangle with vertices $(0, 0)$, $(9, 0)$ and $(0, 12)$ is: 1

(a) 37 **(b)** 35 **(c)** 36 **(d)** 38

7. A linear polynomial $p(x) = ax + b$ has $p(1) = 11$ and $p(2) = 18$. Then $b =$ 1

(a) 3 **(b)** 4 **(c)** 6 **(d)** 5

8. $\sqrt{3} + (5 - \sqrt{3})$ is: 1

(a) imaginary **(b)** rational **(c)** zero **(d)** irrational

9. The value of $78^2 - 22^2$ is: 1

(a) 5599 **(b)** 5602 **(c)** 5600 **(d)** 5601

10. The point $(4, 9)$ is reflected first in the x-axis and then in the y-axis. The final point is: 1

(a) $(-4, -9)$ **(b)** $(4, -9)$ **(c)** $(-4, 9)$ **(d)** $(-9, -4)$

11. The nth term of a pattern is $7n - 2$. Its 12th term is: 1

(a) 82 **(b)** 84 **(c)** 83 **(d)** 81

12. A rational number between $\frac{2}{7}$ and $\frac{4}{7}$ is: 1

(a) $\frac{5}{7}$ **(b)** $\frac{6}{7}$ **(c)** $\frac{1}{7}$ **(d)** $\frac{3}{7}$

13. Factorise $x^2 - 3x - 28$: 1

(a) $(x - 7)(x - 4)$ **(b)** $(x - 6)(x + 5)$ **(c)** $(x + 7)(x - 4)$ **(d)** $(x - 7)(x + 4)$

14. The area of the rectangle with opposite vertices $(1, 2)$ and $(6, 5)$ and sides parallel to the axes is: 1

(a) 14 **(b)** 17 **(c)** 15 **(d)** 16

15. Lines $y = 3x + 2$ and $y = 3x - 7$ are: 1

(a) perpendicular **(b)** intersecting at one point
(c) identical **(d)** parallel

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. For $p(x) = 5x + 1$ and $q(x) = 2x - 8$, find the zero of $p(x) - q(x)$. 2

17. Find the decimal expansion of $\frac{5}{22}$. 2

18. If $x + y = 11$ and $xy = 28$, find $x^2 + y^2$. 2

19. One end of a diameter is $A(9, 4)$ and the centre of the circle is $O(3, -1)$. Find the other end B. 2

20. The nth term of a pattern is $8n - 3$. Find its first three terms and the 10th term. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. If $a - b = 4$ and $ab = 21$, find $a^2 + b^2$ and $(a + b)^2$. 3

22. The vertices of a triangle are $A(0, 0)$, $B(3, 3)$, $C(6, 0)$. Show that it is an isosceles right-angled triangle. 3

23. A pattern has first term 6 and common difference 7. Find its nth term and the term number whose value is 90. 3

24. Without dividing, state whether $\frac{21}{75}$ has a terminating decimal expansion. 3

OR

Factorise $2x^2 + 11x + 12$. 3

25. Show that $A(-2, 0)$, $B(8, 24)$, $C(13, 36)$ are collinear. 3

OR

The line $y = mx + 20$ passes through $(3, 32)$. Find m, then the x-intercept of the line. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Show that the quadrilateral with vertices $A(0, 0)$, $B(5, 1)$, $C(7, 6)$, $D(2, 5)$ is a parallelogram and find the midpoint of the diagonals. 5

OR

A salesperson earns ₹5000 per month plus ₹150 for each item sold. Write her earning $E(n)$, find the earning for 20 items, the number of items sold for ₹14000, and the extra items needed to raise earning by ₹1200. 5

27. Simplify $(3 + \sqrt{5})^2 + (3 - \sqrt{5})^2$ and state whether the result is rational. 5

OR

If $a + b + c = 9$ and $ab + bc + ca = 26$, find $a^2 + b^2 + c^2$. 5

END OF QUESTION PAPER — Total: 50 Marks